BIOS 6611 Midterm Project

Overview

This project is intended to give you practice with conducting a "rigorous" simulation study to examine some question or problem of interest. We have been applying simulations in lecture and homeworks throughout the semester. However, these examples are very targeted and brief. In this project you will work on a simulation problem that you propose, and then write a brief two-page summary report.

For your chosen topic, you'll prepare a 2-page summary (strict maximum; single spaced; at least 1 figure and 1 table must be in the 2-page limit; references can extend beyond the 2-page limit) in the form of a brief scientific report. Page 2 includes an example outline with potential lengths for different sections to consider for your simulation study report.

The rubric that will be used for grading available on the Canvas Midterm Project assignment, and is listed out on Page 3 of this document

Some Potential Ideas:

- Sample size calculation for a study where we do not have an easily accessible power calculator (e.g., non-inferiority study designs, equivalence study designs, risk ratio, chi-squared test, permutation t-test)
- Theorem evaluation (e.g., central limit theorem under various distributions and sample sizes, other theorems you've seen in papers/textbooks/classes)
- Compare methods (e.g., the exact versus asymptotic approaches for a given method, performance of a t-test vs. permutation t-test vs. chi-squared test either for binary data or by coercing a continuous variable, compare asymptotic confidence intervals from a function's output with bootstrap percentile confidence intervals)
- Approximations for distributions (e.g., the Poisson approximation to the binomial)

- Taking any model or approach we've discussed and evaluating what happens when we violate our assumptions (e.g., how poorly does the t-test perform across various measures if we do not have normally distributed data)

A few thoughts on simulation studies:

- Remember, you get to set the "truth" that you are simulating from! You have the power to make things very subtle or extremely drastic.
- When choosing scenarios, it can be helpful to fit one that meets the assumptions that you can compare to (e.g., simulating normal data for a t-test to compare to exponentially simulated data with the same mean). If you are comparing to a known theory (i.e., you have a reference or what should happen theoretically), you may be able to just reference that as your comparator. Although, it can still be useful (and fun) to confirm the theoretical properties via simulation!
- Try to only "break" one part of the problem so you can better attribute why you are observing your results. If you change 3 things at once, it can be hard to know what specific choice on its own led to what you observe.
- The "outcomes" you measure will depend on the question you are exploring via simulation. They may include the mean, median, bias, standard deviation, mean square error, relative efficiency, coverage, power, type I error rate, etc.

Example Outline of Brief Scientific Report/Summary for Simulation Studies

- Abstract – 0.2
- Introduction with brief background – 0.2 page
- Question to explore, motivating problem, theorem description – 0.3 page
- Simulation set-up – 0.3 page
- Results – 1 page
  - At least one plot
  - Tables summarizing either the simulation scenario(s) and/or results
- Discussion and Conclusion – 0.25 page

- References (not included in 2-page limit)
  - You may not have any references for this project and that is okay.
- Appendix for Reproducible Research Goal (not included in 2-page limit):
  - R code with clear comments

Formatting Guidelines

File Formats

- Project should be submitted as pdfs or word documents

Page Limits

- An expected page length of at least 1.5 pages and no more than 2 pages.

Font Size and Spacing

- Use a standard font size and any standard font. Articles should be single spaced.

Abstract and Keywords

- A short abstract should describe the problem, data, and results.
- Immediately after the abstract, provide a maximum of 5 keywords, avoiding general and plural terms and multiple concepts (avoid, for example, 'and', 'of'). These keywords will be used for indexing purposes.

Tables and Figures

- All tables and figures should be "publication ready" with informative titles and labels. Where applicable, units of measurement should be included. Variable names from code should be replaced with meaningful labels (e.g., instead of "outcome_variable4" write the actual outcome).

References

- Any style of reference is acceptable as long as formatting is consistent.

Supplemental Materials

- Given the small page limits, it is acceptable to include *additional* tables and figures beyond the required minimum of one table and one figure in a Supplemental Materials section if needed. These figures and tables should be referenced in the text as **Figure S1**, **Table S1**, etc.

Reproducible Research Principles

- An *appendix* with R, SAS, or other code should be included your references with code that is clearly commented for the simulations, analyses, figures, and tables ultimately included in the report. Code that was ultimately not used in the final report should be excluded from this appendix.
- When grading, we will be running the provided code to verify the results you got within your project. This is a large portion of the overall project grade – so make sure you check that your code runs without error and produces the exact results you provide within the report.

Grading Rubric

Criteria	**Points**	**Details**
Technical Requirements	5	Length (1.5-2 pages), single spacing, etc.
Scientific Writing	5	More formal scientific writing style, not informal or casual writing
Abstract	5	Brief summary of project
Introduction/Background Section	5	Background of simulation project being explored
Motivating Problem/Theorem Explored/Question Examined	5	Clearly state the specific question being examined in your project
Simulation Study Details	10	Detailed description of the simulation approach – should contain enough detail that someone could mimic the simulation by reading it
Results: Table	10	Journal ready table of some component of the results
Results: Figure	10	Journal ready figure of some component of the results

Criteria	Points	Details
Results	5	Summary of the results of the simulation study
Discussion/Conclusion: Interpretation of Results	10	Interpreting the results in context of the original question being examined
Discussion/Conclusion: Limitations	5	Address the limitations of the simulation or the question being examined
Reproducible Research Principles	25	R code for all results clearly presented, with good commenting and seed clearly set. Should be able to easily run and get the same results as the project.

Additional Resources

Several additional resources are available under the Midterm Simulation Project – I have outlined them below:

- A simulation study worksheet:
  - Completely optional but could be a helpful way to organize your thoughts when you are thinking through the topics
- Project Example (The t-test and non-normal data: where did the power go?)
  - One example of what the midterm simulation project report could look like
- RMD shell
  - An Rmarkdown shell that you can use to create your report
  - Note: you don't have to use Rmarkdown for your report – I just provide this to help if you are using it
- Various papers about writing scientific papers/simulations studies
  - Marie Davidian/NCSU simulation study slide deck
  - AJPH example of a brief report
  - Amstat – Tips on writing results for a scientific paper